

Skill Check 3: Building Better Code

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Asking for help on specific questions. (It is fine to ask for clarification!)
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Skill Check Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully. For each two questions correct, you will receive one course point for a total of two course points.

Question 1: FUNCTIONS

Write a function that calculates the sum of two numeric vectors and returns only the biggest of the two sums. Include an example function call to demonstrate the function in operation.

[illegible]

Question 2: LOOPS

Create a while or repeat loop that starts a counter at -10 and prints out the value of the counter divided by 2 until the counter reaches 30.

[illegible]

Question 3: CONDITIONALS

Load in the iris dataset. This dataset contains a column named Species of class factor with 3 levels: setosa, versicolor, and virginica. Write a conditional that changes the Species column to be 0 where Species == "setosa", 1 where Species == "versicolor", and 2 where Species == "virginica".

[illegible]

Question 4: COMBINATIONS

You are performing the following calculation on the results of an ELISA assay of DNA, which is taking the average of densities resulting from duplicate concentrations in a single run:

```
mean_density <- mean(DNase$density[DNase$Run == 1 &
  DNase$conc == unique(DNase$conc)[1]])
mean_density <- mean(DNase$density[DNase$Run == 1 &
  DNase$conc == unique(DNase$conc)[2]])
mean_density <- mean(DNase$density[DNase$Run == 2 &
  DNase$conc == unique(DNase$conc)[1]])
mean_density <- mean(DNase$density[DNase$Run == 2 &
  DNase$conc == unique(DNase$conc)[2]])
```

Write a function that will perform this calculation for all runs and concentrations in the data set. Include a function call that reproduces one of the examples above.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]